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भारतीय मानक मसौदा

चिकित्सा प्रयोगशाला हेतु उपकरण — भंवर मिश्रक — विशिष्टि

Draft Indian Standard

**Medical Laboratory Instruments – Vortex Mixer –
Specifications**

ICS 71.040.20

Medical Laboratory Instruments Sectional Committee,
MHD 10

Last date for comments:
As per portal

FOREWORD

(Formal clauses will be added later)

A vortex mixer is a laboratory device used to mix small vials of liquid. It consists of an electric motor with a vertically oriented drive shaft and a cupped piece mounted slightly off-centre. When the unit is turned on, the cupped piece oscillates rapidly, creating a vortex in the liquid. In medical laboratories, vortex mixers are essential for mixing reagents, suspending cells, or combining solutions for assays. They ensure thorough mixing, enhancing the accuracy and consistency of experiments and diagnostic tests. Their simplicity and effectiveness make them a staple in clinical research.

A national standard for vortex mixers ensures uniformity in performance, safety, and quality across laboratories. This guarantees reliable and reproducible results, minimises equipment failure, and enhances user safety. Standardization also facilitates regulatory compliance and interoperability, promoting consistency in scientific research and diagnostic practices.

To decide whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test or analysis, shall be rounded off per IS 2: 2022 ‘Rules for rounding off numerical values (second revision)’

Draft Indian Standard

**Medical Laboratory Instruments – Vortex Mixer –
Specification**

1 Scope

This Indian Standard specifies the requirements for electrically operated vortex mixers used in laboratories for mechanical mixing and stirring of small volumes of liquids.

2 Normative references

The standards given below contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of these standards.

<i>IS No.</i>	<i>Title</i>
IS 17724 (Part 1):2023	Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use Part 1 General Requirements (IEC 61010-1: 2010 + AMD1:2016 + COR1:2019, MOD)
IS 15575 (Part 1) : 2016 IEC 61672-1 : 2013	Electroacoustics - Sound Level Meters

3 Terms and definitions

For the purposes of this standard, the following terms and definitions apply.

3.1 Base Unit

Main body of the vortex mixer containing the drive mechanism, motor, control system, and supporting structure.

3.2 Cup Head

Interchangeable resilient attachment mounted on the drive mechanism that supports the sample container during mixing.

3.3 Platform Attachment

Interchangeable accessory used for simultaneous mixing of multiple sample containers or microplates.

3.4 Orbit

Diameter of the circular path described by the centre of the cup head or platform during operation.

3.5 Mixing Speed

Rotational or oscillatory operating speed of the drive mechanism, expressed in revolutions per minute (rpm).

3.6 Maximum Sample Capacity

Maximum mass, volume, or number of sample containers that the vortex mixer is designed to operate simultaneously, as declared by the manufacturer.

3.7 Stability

Ability of the vortex mixer to remain stationary without tipping, sliding, or excessive movement during operation under specified conditions.

3.8 Mixing Efficiency

Ability of the vortex mixer to produce a homogeneous mixture within a specified time under prescribed test conditions.

3.9 Duty Cycle

Specified period of operation and rest during which the vortex mixer is intended to operate without exceeding its design limits.

4 General Requirements

4.1 General

4.1.1 The vortex mixer shall be designed and constructed to operate safely and reliably under the environmental and operating conditions specified by the manufacturer.

4.1.2 The instrument shall comply with the applicable safety requirements specified in IS 17724 (Part 1).

4.2 Construction

4.2.1 The vortex mixer shall be of robust construction and shall be free from defects that may affect its performance, safety, or service life.

4.2.2 All components shall be securely assembled and shall withstand the mechanical stresses encountered during normal operation.

4.2.3 The rotating assembly shall be securely attached to prevent loosening or detachment during operation.

4.3 Materials

4.3.1 Materials used for the external housing and components exposed to laboratory reagents shall be resistant to corrosion, cleaning agents, and commonly encountered laboratory chemicals.

4.3.2 Materials shall not exhibit cracking, swelling, deformation, or deterioration under normal conditions of use.

4.4 Stability

4.4.1 The vortex mixer shall remain stable throughout its operation. There shall be no tipping, no detachment of components, and no movement sufficient to impair safe operation or intended performance.

4.5 Controls

4.5.1 The operating controls shall be clearly identified and easily accessible.

4.5.2 Adjustment of operating parameters shall not result in unsafe operation.

4.5.3 Where provided, the direction of adjustment shall be clearly indicated.

4.6 Operating Modes

4.6.1 The vortex mixer shall provide touch operation, continuous operation, or both, as declared by the manufacturer.

4.7 Accessory Mounting

4.7.1 Cup heads, platform attachments, and other accessories supplied by the manufacturer shall be securely fitted and shall not become detached during operation.

4.8 Electrical Supply

4.8.1 The rated voltage, frequency, and rated input shall be declared by the manufacturer.

4.9 Protection Against Overheating

4.9.1 The vortex mixer shall comply with the applicable requirements for equipment temperature limits and resistance to heat specified in Clause 10 of IS 17724 (Part 1).

4.10 Finish

4.10.1 Metallic parts shall be adequately protected against corrosion.

4.10.2 The external finish shall withstand normal laboratory cleaning and handling without deterioration.

4.11 Noise and Vibration

4.11.1 The instrument shall operate without abnormal vibration, excessive noise, or mechanical instability.

5. Construction requirements

5.1 Housing

The housing shall enclose the internal components necessary for safe operation and shall provide protection against access to hazardous parts and mechanical damage during normal use.

5.2 Drive Mechanism

The drive mechanism shall produce the declared orbital motion without abnormal vibration or mechanical interference.

5.3 Accessories

Cup heads and platform attachments shall fit securely and be readily removable without damage.

5.4 Control System

The speed control shall permit adjustment of the mixing speed throughout the declared operating range without interruption of operation or unintended stopping.

6. Performance requirements

6.1 Speed Accuracy

Measured speed shall not differ from the declared speed by more than ± 10 percent.

6.2 Orbit Diameter

Measured orbit diameter shall not differ from the declared value by more than ± 0.5 mm.

6.3 Mixing Efficiency

The vortex mixer shall produce homogeneous mixing under the specified test conditions when tested in accordance with 8.5.

6.4 Endurance

The vortex mixer shall operate continuously for **8 h** at the maximum declared speed without failure.

6.5 Noise

Noise level shall not exceed 70 dB(A) measured at 1 m.

7. Safety requirements

7.1 Electrical safety shall comply with IS 17724 (Part 1).

7.2 Protective earthing where applicable, shall comply with IS 17724 (Part 1).

7.3 Moving parts shall not present mechanical hazards during normal operation.

8 Testing requirements

8.1 Visual Examination

8.1.1 The vortex mixer shall be visually examined for compliance with the requirements specified in Clauses 4.2, 4.3, 4.5, 4.7 and 4.10.

8.1.2 The examination shall include verification of:

- a) workmanship,
- b) finish,
- c) assembly of components,
- d) identification of controls,
- e) accessory mounting and
- f) visible defects.

8.1.3 The vortex mixer shall comply with the applicable requirements specified in Clauses 4.2, 4.3, 4.5, 4.7.

8.1.4 Compliance with 4.10.1 shall be verified by visual examination. The exposed metallic parts shall be free from visible corrosion, flaking or deterioration that may affect the safety or intended performance of the vortex mixer.

8.1.5 Compliance with 5.1 shall be verified by visual examination and by the applicable enclosure and mechanical strength tests specified in IS 17724 (Part 1).

8.2 Speed Accuracy Test

8.2.1 Apparatus

A calibrated non-contact tachometer having an accuracy of ± 0.5 percent or more.

8.2.2 Procedure

The speed of the vortex mixer shall be measured at the minimum, intermediate and maximum settings declared by the manufacturer. The measured speed shall be recorded after steady operation has been achieved.

8.2.3 Acceptance Criteria

The measured speed shall comply with the requirements specified in **6.1**.

8.3 Orbit Diameter Measurement

8.3.1 Apparatus

Suitable measuring instrument having a least count of 0.1 mm.

8.3.2 Procedure

The orbit diameter shall be measured as the diameter of the circular path described by the centre of the cup head during operation.

8.3.3 Acceptance Criteria

The measured orbit diameter shall comply with the requirements specified in **6.2**.

8.4 Stability Test

8.4.1 Procedure

- a) The vortex mixer shall be placed on a rigid, horizontal surface,
- b) The standard cup head shall be fitted,
- c) The mixer shall be operated in the continuous mode at the maximum declared speed for 30 min,
- d) The instrument shall be observed throughout the test.

8.4.2 Acceptance Criteria

During and after completion of the test,

- a) the vortex mixer shall not tip over,
- b) no component shall become detached,
- c) no movement shall occur that impairs safe operation and
- d) the vortex mixer shall remain functional.

8.5 Mixing Performance Test

8.5.1 Apparatus

- a) 50 ml centrifuge tube,
- b) distilled water,
- c) aqueous dye solution.

8.5.2 Procedure

- a) Fill the centrifuge tube with 50 ml of distilled water.
- b) Add 0.5 ml of dye solution.
- c) Operate the vortex mixer at the maximum declared speed for 10 s.
- d) Visually examine the contents immediately after mixing.

8.5.3 Acceptance Criteria

No visible streaks, sediment, or unmixed regions shall be observed.

8.6 Temperature Rise Test

8.6.1 The temperature rise of the vortex mixer shall be determined in accordance with the applicable requirements specified in IS 17724 (Part 1).

8.6.2 The measured temperature rise shall comply with the requirements specified in IS 17724 (Part 1).

8.7 Noise Measurement

8.7.1 Apparatus

A sound level meter conforming to class 2 as specified in IS 15575 (Part 1).

8.7.2 Procedure

The sound pressure level shall be measured at a distance of 1 m from the vortex mixer while operating at the maximum declared speed.

8.7.3 Acceptance Criteria

The measured sound pressure level shall comply with the requirements specified in Clause 6.5.

8.8 Endurance Test

8.8.1 Procedure

The vortex mixer shall be operated continuously at the maximum declared speed for 8 h under no-load conditions. Following completion of the test, the instrument shall be visually examined and operated.

8.8.2 Acceptance Criteria

After completion of the test,

- a) the vortex mixer shall remain functional,
- b) no component shall become loose or detached,
- c) no abnormal vibration or noise shall be observed and
- d) the vortex mixer shall comply with the applicable performance requirements specified in 6.

9. Marking

9.1 The instruments shall be legibly and indelibly marked with the following:

- a) The name of the manufacturer, initials or trademark,
- b) batch or lot number,
- c) the date of manufacture,
- d) maximum speed of vortex mixer,
- e) Country of manufacture

9.2 BIS CERTIFICATION MARKING

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the Bureau of Indian Standard Act, 2016 and the Rules and Regulations framed thereunder, and the product(s) may be marked with the Standard Mark.